

Solutions to JEE Advanced Booster Test – 1 | 2022 | Code A

[PHYSICS]

1.(BD) For m_1 : $F_1 - T = \frac{m_1(F_1 - F_2)}{m_1 + m_2}$

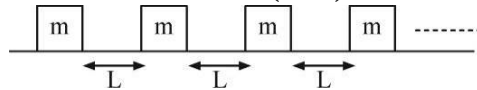
$$\Rightarrow T = \frac{m_2 F_1 + m_1 F_2}{m_1 + m_2}$$

If $m_1 \gg m_2$, $T \approx F_2$

If $m_2 \gg m_1$, $T \approx F_1$

If $m_1 = m_2$, $T = \frac{F_1 + F_2}{2}$

2.(ABC) Total no. of collision = $(n - 1)$



For inelastic collisions,

Let $V_1, V_2, V_3, \dots$ be velocities after 1, 2, 3, $\dots$ collisions.

$$\therefore mV = 2mV_1 = 3mV_2 = \dots$$

$$\Rightarrow V_1 = \frac{V}{2}, V_2 = \frac{V}{3}, V_3 = \frac{V}{4}, \dots \quad \therefore t_1 = \frac{L}{V}, t_2 = \frac{2L}{V}, t_3 = \frac{3L}{V}, \dots$$

Total time elapsed before last collision

$$= t_1 + t_2 + \dots + t_{n-1} = \frac{L}{V}(1 + 2 + 3 + \dots + n-1) = \frac{n(n-1)}{2} \frac{L}{V}$$

Applying momentum conservation between initial and final gives $mV = nmV_f \Rightarrow V_f = \frac{V}{n}$

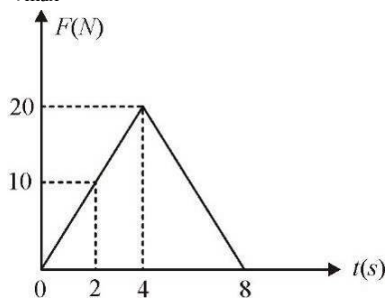
For elastic collisions, velocities keep getting exchanged.

All collisions take equal times $t = \frac{L}{V} \quad \therefore$ Total time $(n-1) \frac{L}{V}$

Finally, the last block is moving with speed V and all other blocks are at rest. So $V_{cm} = \frac{mV}{nm} = \frac{V}{n}$

3.(ABC) $\mu = 0.25$

$$f_{\max} = 0.25 \times 4 \times 10 = 10 \text{ N}$$



Block will start moving at 2s

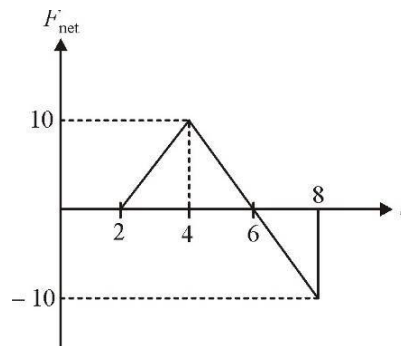
$$F_{\text{net}} = F_{\text{applied}} - \text{friction}$$

Net impulse on the ball

From 2s to 8s

$$= \frac{1}{2} \times 2 \times 10 = 10 \text{ Ns} = \text{change in momentum}$$

Velocity of the mass at $t = 8\text{s}$ is $10/4 = 2.5 \text{ m/s}$



Block has maximum velocity at $t = 6s$

Net impulse on the ball from 2s to 6s = 20 Ns

Velocity of ball at ($t = 6s$) = 5 m/s. After $t = 8s$ a constant retarding friction force of 10N acts and the block has momentum

= 10Ns at $t = 8s$.

Box will come to rest when impulse of 10Ns acts on it in the opposite direction of its motion

Box will come to rest 1s after $t = 8s$, i.e. at $t = 9s$.

$$4.(AD) \quad U(x) = \frac{U_0}{2} \left[\frac{x}{a} - \frac{1}{3} \left(\frac{x}{a} \right)^3 \right]$$

$$F(x) = -\frac{dU}{dx} = -\frac{U_0}{2} \left[\frac{1}{a} - \frac{x^2}{a^3} \right] = \frac{U_0}{2a} \left[\frac{x^2}{a^2} - 1 \right]$$

Force acting on the particle is zero at $x = \pm a$

$$\frac{d^2U}{dx^2} = \frac{U_0}{2} \left[-\frac{2x}{a^3} \right] = -\frac{U_0}{a^3} x$$

For $x = -a$

$$\frac{d^2U}{dx^2} > 0, \text{ it is a position of stable equilibrium}$$

For $x = +a$

$$\frac{d^2U}{dx^2} < 0, \text{ it is a position of unstable equilibrium}$$

$$5.(ACD) \quad AB = 2R \cos \theta$$

$$AB = \frac{1}{2} g \cos \theta t^2 \Rightarrow 2R \cos \theta = \frac{1}{2} g \cos \theta t^2$$

$$2\sqrt{\frac{R}{g}} = t; \quad v = \sqrt{2gh}$$

$$6.(BCD) \quad \text{Take friction up the plane and put } f = \mu N \text{ to get : } F_{\min} = mg \sin \theta - \mu mg \cos \theta$$

$$\text{Take friction down the plane and put } f = \mu N \text{ to get : } F_{\max} = mg \sin \theta + \mu mg \cos \theta$$

$$\Rightarrow \text{for no sliding, } mg \sin \theta - \mu mg \cos \theta < F < mg \sin \theta + \mu mg \cos \theta$$

$$\text{Put } f = 0 \text{ to get } F = mg \sin \theta$$

$$7.(B) \quad \text{Time after which block will move}$$

$$F \cos \theta \geq \mu_s N$$

$$F \cos \theta \geq \mu_s (mg - F \sin \theta)$$

$$F \geq \frac{\mu_s mg}{(\cos \theta + \mu_s \sin \theta)}$$

$$t \geq \frac{\mu_s mg}{(\cos \theta + \mu_s \sin \theta) \alpha}$$

$$t_{\min} = \frac{\mu_s mg}{\alpha \sqrt{1 + \mu_s^2}} = 2 \text{ sec}$$

$$\text{When } \theta = 37^\circ$$

8.(C) For $\theta = 37^\circ$, after $t = 2\text{sec}$

$$N = 10 - \frac{9t}{5}$$

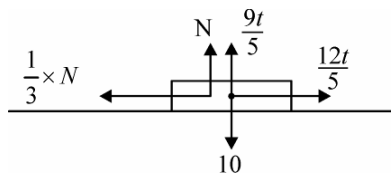
Block leaves contact when $N = 0$

$$\Rightarrow 9t/5 = 10$$

$$t = \frac{50}{9}\text{s}$$

Impulse = $\Delta p = mv$

$$\Rightarrow v = \int_2^{50/9} \left(\frac{12t}{5} - \frac{N}{3} \right) dt = \int_2^{50/9} \left(3t - \frac{10}{3} \right) dt \approx 28\text{ m/s}$$



9.(A) When the length of hanging part is x , the spring is elongated by x

$$\begin{aligned} \therefore a &= \frac{\lambda x g - kx}{\lambda l} \\ &= \frac{100x - 50x}{10} \quad (\text{as } \lambda = 0.1\text{ kg/cm} = 10\text{ kg/m}) \\ &= 5x \end{aligned}$$

10.(B) $a = 5x \Rightarrow \frac{v dv}{dx} = 5x$

$$\int_0^v v dv = \int_0^x 5x dx$$

$$\frac{v^2}{2} = \frac{5x^2}{2} \Rightarrow v = \sqrt{5}x$$

Or by energy conservation

loss in GPE = Increase in EPE + Increase in KE

$$\lambda x g \left(\frac{x}{2} \right) = \frac{1}{2} kx^2 + \frac{1}{2} \lambda l v^2$$

$$\text{So } 50x^2 = 25x^2 + 5v^2 \Rightarrow v^2 = 5x^2$$

11.(B) $r = l \sin \theta$

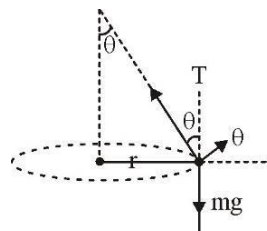
$$= 0.5 \times \frac{3}{5} = 0.3 \quad \rightarrow T \cos \theta = mg$$

$$T \times \frac{4}{5} = 20 \Rightarrow T = 25\text{ N}$$

$$\rightarrow T \sin \theta = ma \Rightarrow a = \frac{25}{2} \times \frac{3}{5} = \frac{15}{2} \text{ m/s}^2 = a$$

$$\rightarrow w^2 r = a \Rightarrow w = \sqrt{\frac{15}{2 \times 0.3}} = \sqrt{\frac{50}{2}} = 5 \text{ rad/s}$$

$$\rightarrow v = wr \Rightarrow v = 5 \times 0.3 = \frac{3}{2} \text{ m/s}$$



12.(C) $mg l(1 - \cos \theta) = \frac{1}{2} mv^2$

$$v = \sqrt{2gl(1 - \cos \theta)}$$

For the particle to complete the full circle

$$v = \sqrt{2gl(1 - \cos \theta)} \geq \sqrt{5g(l - d)}$$

$$2l(1 - \cos \theta) \geq 5l - 5d$$

$$d \geq \frac{l}{5}(3 + 2 \cos \theta)$$

For $\theta = 60^\circ$ $d \geq \frac{l}{5}\left(3 + 2 \times \frac{1}{2}\right) = \frac{4l}{5} = \frac{40l}{50}$

For $\theta = 53^\circ$ $d \geq \frac{l}{5}\left(3 + 2 \times \frac{3}{5}\right) = \frac{42l}{50}$

For $\theta = 60^\circ$ and $d = 0.6l$

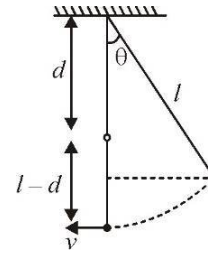
$$v_{\text{bottom}} = \sqrt{2gl(1 - \cos \theta)} = \sqrt{2 \times 10 \times l \left(1 - \frac{1}{2}\right)} = \sqrt{10l} = \sqrt{gl}$$

Radius of circle centered around nail = $0.4l$

$$T_{\text{string}} = \frac{mv^2}{l - d} + mg = \frac{m \times gl}{0.4l} + mg = mg(2.5 + 1) = 3.5mg$$

For $\theta = 53^\circ$ and $d = 0.6l$

$$T_{\text{string}} = \frac{mv^2}{l - d} + mg = \frac{m \times \left[\sqrt{2 \times gl \times \left(1 - \frac{3}{5}\right)} \right]^2}{0.4l} + mg = mg \left[\frac{4/5}{0.4} + 1 \right] = 3mg$$



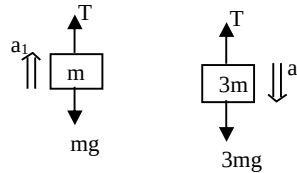
13.(2.50) Let a_1 be acceleration of the blocks

$$3mg - T = 3ma_1$$

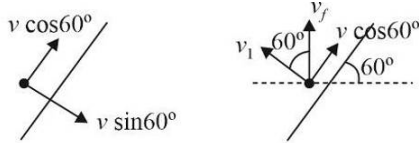
$$T - mg = ma_1$$

Solving $a_1 = \frac{g}{2}$

$$a_{\text{cm}} = \frac{3ma_1 - ma_1}{4m} = \frac{a_1}{2} = \frac{g}{4} = 2.5 \text{ m/s}^2$$



14.(0.33)



Just before collision

Just after collision

$$\frac{v_1}{v \sin 60^\circ} = e \Rightarrow v_1 = ev \sin 60^\circ$$

As final velocity is directed vertically upwards,

$$\tan 60^\circ = \frac{v \cos 60^\circ}{ev \sin 60^\circ} \Rightarrow e = \frac{1}{3}$$

15.(5) $0 = \Delta x_{1G} + 2\Delta x_{2G} + 3\Delta x_{3G} = (\Delta x_{13} + \Delta x_{3G}) + 2(\Delta x_{23} + \Delta x_{3G}) + 3\Delta x_{3G} = -30 + 2(30) + 6\Delta x_{3G}$

$$\therefore \Delta x_{3G} = -\frac{30}{6} = -5 \text{ cm}$$

16.(40) $\vec{F} = 3xy^2\hat{i} + y\hat{j}$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$dw = \vec{F} \cdot d\vec{r} = 3xy^2 dx + y dy = 3x \cdot x^4 dx + y dy$$

$$w = \int_0^2 3x^5 dx + \int_0^4 y dy = \frac{3}{6} [x^6]_0^2 + \left[\frac{y^2}{2} \right]_0^4 = 32 + 8 = 40$$

17.(5) $P = \frac{dw}{dt}$

$$\int P dt = \int dw$$

$$0.5 \times 5 = \frac{1}{2} \times 0.2 V^2 - 0 \quad \left\{ \int dw = \Delta KE \right\}$$

$$2.5 = 0.1 V^2$$

$$V = 5 \text{ m/s}$$

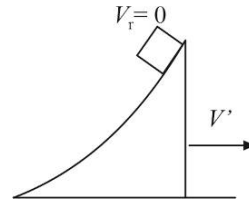
18.(10) To reach the top of wedge, the velocity of block relative to wedge shouldn't become zero before reaching the top. For $V_{\min}$, relative speed at top should be zero.

Momentum conservation $1 V = 5 V' \Rightarrow V' = \frac{V}{5}$

Energy conservation

$$\frac{1}{2} (1) V^2 = 1(10)(4) + \frac{1}{2} (5) \frac{V^2}{25}$$

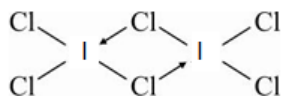
$$V^2 - \frac{V^2}{5} = 80 \Rightarrow V^2 = 100 \Rightarrow V = 10 \text{ m/s}$$



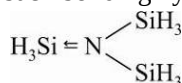
[CHEMISTRY]

- 1.(ABCD) (A) Total electron pairs around nitrogen in NO_4^+ is four hence hybridization state is sp^3 . N can form maximum four bonds.
- (B) Refer NCERT
- (C) Molecular shape of XeF_5^- is pentagonal planar due to the presence of two lone pair of electrons.
- (D) $\begin{array}{c} \text{F} \\ \diagup \\ \text{F} \rightarrow \text{S} \equiv \text{N} \\ \diagdown \\ \text{F} \end{array}$ Hybridization state of S is sp^3 .
- 2.(BD) In presence of Phenolphthalein; only Na_2CO_3 will react with HCl and on reaction with HCl, Na_2CO_3 converts to NaHCO_3 . Hence correct choices are (B), (D)

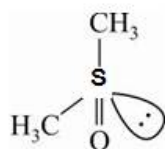
- 3.(CD) (A) I_2Cl_6 is dimer of ICl_3 and its structure is as shown and geometry of I is square planar.



- (B) Due to back bonding hybridization state of N is sp^2 not sp^3 .



- (C) Nitrogen will be sp^3 hybridization if it forms four σ bonds.
- (D) Sulphur is sp^3 hybridization due to lone pair of electron.



4. (CD)

- 5.(ABCD)
- | Molecular species | C_2 | C_2^{2-} | NO | NO^+ | O_2 | O_2^- |
|-------------------|--------------|-------------------|-------------|---------------|--------------|----------------|
| Bond order | 2 | 3 | 2.5 | 3 | 2 | 1.5 |
| Magnetic property | Diamagnetic | | Diamagnetic | Paramagnetic | Diamagnetic | |
| Paramagnetic | Paramagnetic | | | | | |
- In C_2 no σ bond while in C_2^{2-} there is one σ bond

- 6.(ABCD) At higher temperature more number of molecules have higher velocity and curve becomes broader but area under the curve remains same.

- 7.(C) $\text{P} \rightarrow 0.1 \text{ M HCl, } 50 \text{ ml} \longrightarrow 5 \text{ meq of } \text{H}^+ \text{ in } 50 \text{ ml}$ now 25 ml, 0.1 M NaOH is added

So, $\text{Q} \rightarrow 2.5 \text{ meq of } \text{H}^+ \text{ in } 75 \text{ ml solution}$ now 25 ml H_2O is further added

$\text{R} \rightarrow 2.5 \text{ meq of } \text{H}^+ \text{ in } 100 \text{ ml solution}$

25 ml, 0.1 M HCl and 25 ml H_2O added

$\therefore \text{S} \rightarrow 5 \text{ meq of } \text{H}^+ \text{ in } 150 \text{ ml solution}$

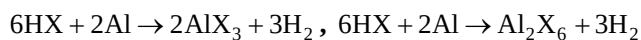
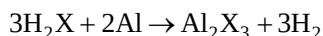
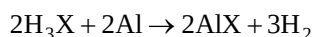
50 ml of 0.1 M $\text{Ca}(\text{OH})_2$ added

So, $\text{T} \rightarrow 5 \text{ meq of } \text{H}^+ \text{ and } 10 \text{ meq of } \text{OH}^- \Rightarrow 5 \text{ meq of } \text{OH}^- \text{ in } 200 \text{ ml}$

In 40 ml of T $\rightarrow 1.0 \text{ meq of } \text{OH}^-$

5 ml of P $\rightarrow 0.5 \text{ meq of } \text{H}^+$

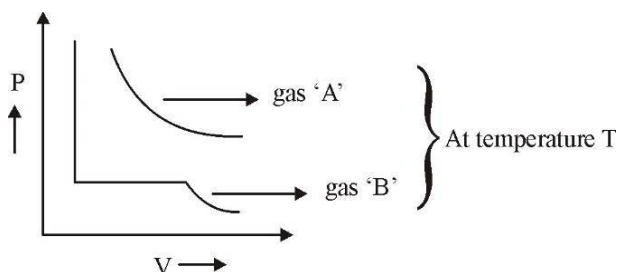
$\therefore$ 40 ml of T can not neutralize completely 5 ml of P

8.(D) For $n=1$ For $n=2$ For $n=3$ 

For all the above cases

2 mol of Al react to produce 3 mol of H_2

9.(D)



Gas A acts as an ideal gas while gas B acts as van der waal's gas and temperature T is lower than its critical temperature hence it undergoes liquefaction and shape of isotherms will change. Boyle's

temperature for a gas is $T_B = \frac{a}{Rb}$ while critical temperature is $T_C = \frac{8a}{27Rb}$

10.(C) At temperature greater than T_C of gas 'B', curve will take same shape as gas 'A'. But at temperature $T > T_C$, gas 'A' curve shape will remain same

At temperature T, gas 'B' can be liquified implies intermolecular attractive forces dominating over random motion, therefore, $\frac{an^2}{V^2}$ is less negligible w.r.t. pressure.

11.(B) [(I)-q, r]; [(II)-p, r, t]; [(III)- t]; [(IV)-r, t]

C_2H_4	2c -2e bonds,	B_2H_6	3c -2e bonds,
	Molecular plane is nodal plane for π -bond,		No π -bond,
	Non-polar molecule		Non-polar molecule
	No coordinate bond		No coordinate bond
	planar molecule,		Non-planar molecule,
N_2H_4	2c -2e bonds,	Al_2Cl_6	2c -2e bonds,
	No π -bond,		No π -bond,
	polar molecule		Non-polar molecule
	No coordinate bond		coordinate bond
	Non-planar molecule,		Non-planar molecule,

12.(B) [(I)-r, t]

[(II)-q]

[(III)-s]

[(IV)-p]

	Column 1	Column 2
(I)	If only force of attraction among the gas molecules be negligible	$PV = RT + Pb$, $V_m > 22.4$ (at STP)
(II)	If only the volume of the gas molecules be negligible	$PV = RT - \frac{a}{V}$
(III)	If both force of attraction and volume of gas molecules is negligible	$PV = RT$
(IV)	If both the force of attraction and volume of gas molecule is significant	$\left(P + \frac{a}{V^2}\right)(V - b) = RT$

13.(9) Be_2^{2-} , B_2 , C_2 , N_2 , O_2 , F_2^{2+} , N_2^+ , CN^- , N_2^{2+} all possess pi bonds.

14.(26) $X_1 = 4$, $X_2 = 9$, $X_3 = 9$, $X_4 = 4$,

15.(400) volume $\times$ density = mass,

$$\text{Number of mole} = \frac{\text{mass}}{\text{molecular mass}}$$

$$\text{Mass} = 10 \times 32 \Rightarrow 320 \text{ g}$$

$$\text{Volume} = \frac{\text{Mass}}{\text{Density}} = \frac{320}{0.8} = 400 \text{ mL}$$

$$16.(11.72) \quad PV = Z nRT, \quad w = \frac{PVM}{ZRT} = \frac{96 \times 0.082 \times 32}{0.96 \times 0.082 \times 273} = 11.721 = 11.72g$$

17.(35.50) Cl_2 has the smallest mean free path because as the radius of the molecule increases the space between the molecules decreases causing the number of collisions to increase, thus decreasing the mean free path.

18.(0.87) Gas 1 : Molecular mass M : ratio 2 : 1
Gas 2 : Molecular mass 2M : ratio 2 : 1

$$\text{Molecular mass of mixture} = \frac{2M + 2M}{3} = \frac{4M}{3}$$

$$\therefore \frac{r_{\text{mix}}}{r_{\text{gas A}}} = \sqrt{\frac{M}{4M/3}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2} = 0.866 = 0.87$$

[MATHEMATICS]

$$\begin{aligned}
 1.(CD) \text{ Here, } \frac{T_6 - T_4}{T_6} &= \frac{-(T_4 - T_6)}{T_6} = - \left[\frac{\sin^4 \theta - \sin^6 \theta + \cos^4 \theta - \cos^6 \theta}{\sin^6 \theta + \cos^6 \theta} \right] \\
 &= - \left[\frac{\sin^4 \theta \cos^2 \theta + \cos^4 \theta \sin^2 \theta}{(\sin^2 \theta + \cos^2 \theta)^3 - 3 \sin^2 \theta \cos^2 \theta (\sin^2 \theta + \cos^2 \theta)} \right] \\
 &= - \left[\frac{\sin^2 \theta \cos^2 \theta}{1 - 3 \sin^2 \theta \cos^2 \theta} \right] = - \left[\frac{\sin^2 2\theta}{4 - 3 \sin^2 2\theta} \right] = m \\
 \Rightarrow \sin^2 2\theta &= \frac{4m}{-1 + 3m} \quad \therefore \quad 0 \leq \frac{4m}{-1 + 3m} \leq 1 \quad \Rightarrow \quad m \in [-1, 0]
 \end{aligned}$$

Hence, (C) and (D) are correct answers.

$$\begin{aligned}
 2.(ABC) \quad K &= -\sin^4 x - \cos^4 x - \sin 2x \\
 &= -(1 - 2 \sin^2 x \cos^2 x) - \sin 2x = -1 + \frac{1}{2} \sin^2 2x - \sin 2x = \frac{1}{2} (\sin 2x - 1)^2 - \frac{3}{2} \in \left[-\frac{3}{2}, \frac{1}{2} \right]
 \end{aligned}$$

3.(BCD) We can partition A into 4 subsets L_0, L_1, L_2, L_3 where elements in L_0 are perfectly divisible by 4, elements in L_1 leaves remainder 1, elements in L_2 leave remainder 2 & elements in L_3 leaves remainder 3. There are 500 numbers in each L_1, L_2, L_3 but 499 numbers in L_0 . For the set L_1 , arrange all the numbers in an ascending order and consider the 250 pairs formed by its numbers (1, 5), (9, 13), (17, 21) (1993, 1997). If more than 250 numbers in L_1 are selected out as part of L, then there must be some two numbers coming from one of the above pairs, so their difference is 4. Thus, any more than 250 numbers in L_1 can't be put in L. However, all numbers on odd numbered places or all numbers on even places can be chosen. Thus, at most 250 numbers can be chosen from L_1 as a subset of L, and the same analysis works for each of L_2, L_3 & L_0 (in L_0 all the 250 numbers which are odd multiples of 4 can be chosen to put in L). Therefore, the maximum number of elements in L is 1000.

$$\begin{array}{ccccc}
 4.(BC) & R_1 & R_2 & R_3 & R_4 & R_5 \\
 & 2 & 1 & 1 & 1 & 1 \\
 & 1 & 1 & 2 & 1 & 1 \\
 & 1 & 1 & 1 & 1 & 2
 \end{array}$$

Total ways

$$\begin{aligned}
 & {}^3C_2 \times {}^3C_1 \times {}^3C_1 = 27 \\
 & {}^3C_1 \times {}^3C_2 \times {}^3C_1 = 27 \\
 & {}^3C_1 \times {}^3C_1 \times {}^3C_2 = 27 \\
 & = 81 \left(\frac{6!}{2! 3!} \right) = 4860
 \end{aligned}$$

$$\begin{aligned}
 5.(ABD) \quad x &= \cos 39^\circ + \cos 33^\circ = 2 \cos 36^\circ \cos 3^\circ \\
 y &= \cos 40^\circ + \cos 32^\circ = 2 \cos 36^\circ \cos 4^\circ \\
 z &= \cos 41^\circ + \cos 31^\circ = 2 \cos 36^\circ \cos 5^\circ \\
 \therefore x &> y > z \\
 \text{Since } x &> y > z > 0, \text{ therefore } x + z > y.
 \end{aligned}$$

6.(ABC) THING, NIGHT

GHINT Total Number of words = 120

G ---- 4! = 24 words

H ---- 4! = 24 words

I ---- 4! = 24 words

NG --- 3! = 6 words

NH --- 3! = 6 words

NIGHT 85th words

N ---- 4! = 24 words

TG --- 3! = 6 words

THG -- 2! = 2 words

THIGN 4! × 4 + 6 + 2 + 1 = 105

THIGN 106th words

k = 20 words in between

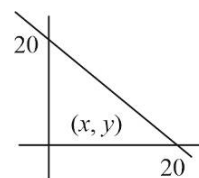
$$= 2^2 \times 5^1$$

$$x + y < 20$$

$$x + y + z = 20$$

$$x, y, z \geq 1$$

$$\text{Number of solutions} = {}^{19}C_2 = 171$$



7.(C) Number of ways = ${}^2C_1 \times {}^2C_1 \times 2! \times 6!$ {select an Indian & American Husband and they can be permuted around the unmarried person in 2! ways}

8.(D) Total arrangements without any constraint = 9!

Relative positions of a Husband & a wife = 2!

Since there are four couples the relative positions of a Husband & a Wife are $(2!)^4$

Since every Wife must appear for her interview only after her Husband,

$$\text{Total number of desired arrangements} = \frac{9!}{(2!)^4}$$

9.(D) $\cos^7 x = 1 - \sin^4 x$

$$\cos^7 x = (1 - \sin^2 x)(1 + \sin^4 x)$$

$$\cos^7 x = \cos^2 x(1 + \sin^2 x)$$

$$\cos^2 x(\cos^5 x - (1 + \sin^2 x)) = 0$$

$$\cos^2 x = 0 \text{ or } \cos^5 x = 1 + \sin^2 x$$

$$x = (2n+1)\frac{\pi}{2} \left| \begin{array}{l} \cos x = 1 \text{ \& } \sin x = 0 \\ x = 2n\pi \end{array} \right.$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \left| \begin{array}{l} x = 0, 2\pi \end{array} \right.$$

10.(D) $\cos(p \sin x) = \sin(p \cos x)$

$$\therefore \cos \theta = \sin \phi \text{ only when } \theta = \phi = \frac{\pi}{4} \text{ or } \theta = \phi = \frac{5\pi}{4}$$

$$\therefore p \sin x = \frac{\pi}{4} \text{ or } p \sin x = \frac{5\pi}{4}$$

$$\therefore p \cos x = \frac{\pi}{4} \text{ or } p \cos x = \frac{5\pi}{4}$$

$$\text{Squaring and adding we get } p^2 = \frac{2\pi^2}{16} \text{ or } p^2 = 2\left(\frac{5\pi}{4}\right)^2$$

$$\text{Smallest positive number } \Rightarrow p = \frac{\sqrt{2}\pi}{4} = \frac{\pi}{2\sqrt{2}}$$

11.(B) [A-q] [B-p] [C-s] [D-p]

(A) We have, $4\cos x(2-3\sin^2 x) + (\cos 2x+1) = 0$

$$\Rightarrow 4\cos x(3\cos^2 x - 1) + 2\cos^2 x = 0 \Rightarrow 2\cos x(6\cos^2 x + \cos x - 2) = 0$$

$$\Rightarrow 2\cos x(3\cos x + 2)(2\cos x - 1) = 0 \Rightarrow \text{either } \cos x = 0 \text{ which gives } x = \pi/2$$

$$\text{or, } \cos x = -2/3,$$

$$\text{Which gives no value of } x \text{ for which } 0 \leq x \leq \pi/2 \text{ or } \cos x = 1/2, \text{ which gives } x = \pi/3$$

$$\text{So, the required difference} = \pi/2 - \pi/3 = \pi/6.$$

(B) Since $2\cos x, |\cos x|$ and $1-3\cos^2 x$ are in G.P.

$$\therefore \cos^2 x = 2\cos x(1-3\cos^2 x)$$

$$\Rightarrow 6\cos^3 x + \cos^2 x - 2\cos x = 0 \Rightarrow \cos x(2\cos x - 1)(3\cos x + 2) = 0$$

$$\Rightarrow \cos x = 0, \frac{1}{2}, -\frac{2}{3}, \cos x \neq 0$$

(C) We have, $\sin x\sqrt{8\cos^2 x} = 1 \Rightarrow \sin x|\cos x| = \frac{1}{2\sqrt{2}}$

Case I: when $\cos x > 0$

$$\text{In this case, } \sin x \cos x = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \sin 2x = \frac{1}{\sqrt{2}} \Rightarrow 2x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4} \Rightarrow x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{9\pi}{8}, \frac{11\pi}{8}$$

As x lies between 0 and 2π and $\cos x > 0$

$$\therefore x = \frac{\pi}{8}, \frac{3\pi}{8}$$

Case II: when $\cos x < 0$

$$\text{In this case } \sin x \cos x = \frac{-1}{2\sqrt{2}}$$

$$\Rightarrow \sin 2x = \frac{-1}{\sqrt{2}} \Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8} \Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8} \text{ as } \cos x < 0$$

Thus, the values of x satisfying the given equation which lie between 0 and 2π are :

$$\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$$

These are in A.P. with common difference $\frac{\pi}{4}$.

(D) We know that the maximum values of $|\sin \theta + \cos \theta|$ and $1 + \sin 2\theta$ are $\sqrt{2}$ and 2, respectively.

$$\text{Also, } (\sqrt{2})^2 = 2$$

$\therefore$ The given equation can hold only if $\sin \theta + \cos \theta = \pm\sqrt{2}$ and $1 + \sin 2\theta = 2$

$$\text{Now, } \sin \theta + \cos \theta = \pm\sqrt{2} \Rightarrow \cos\left(\theta - \frac{\pi}{4}\right) = \pm 1 \Rightarrow \theta = n\pi + \frac{\pi}{4}$$

$$\text{Also, } 1 + \sin 2\theta = 2 \Rightarrow \sin 2\theta = 1 \Rightarrow \theta = \frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$$

Thus, the values of θ in $[-\pi, \pi]$, satisfying both the equation are $\frac{\pi}{4}, -\frac{3\pi}{4}$

12.(B) **INDIANOIL**

- (A) **INDIA NOIL** $5!$
- (B) $I \text{ --- } I$ $\frac{7!}{2!} = 21 \times 5!$
- (C) $\underbrace{\text{---}}_{D L N N} \underbrace{\text{---}}_{I I I A O}$ $\frac{4!}{2!} \times \frac{5!}{3!} = 2 \times 5!$
- (D) ---
1 3 5 7 9 $\frac{4!}{2!} \times \frac{5!}{3!} = 2 \times 5!$

13.(2.67) $|4 \sin x + \sqrt{2}| < \sqrt{6}$

$$-\sqrt{6} < 4 \sin x + \sqrt{2} < \sqrt{6}$$

$$-\sqrt{6} - \sqrt{2} < 4 \sin x < \sqrt{6} - \sqrt{2}$$

$$\frac{-\sqrt{6} - \sqrt{2}}{4} < \sin x < \frac{\sqrt{6} - \sqrt{2}}{4}$$

$$\frac{-\sqrt{2}(\sqrt{3}+1)}{4} < \sin x < \frac{\sqrt{2}(\sqrt{3}-1)}{4}$$

$$\frac{-5\pi}{12} < x < \frac{\pi}{12} \quad \text{for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ Comparing with } \frac{a\pi}{24} < x < \frac{b\pi}{24}$$

We get $a = -10, b = 2$

$$\therefore \left| \frac{a+b}{3} \right| = \left| \frac{-10+2}{3} \right| = \left| \frac{-8}{3} \right| = 2.67$$

14.(7.69) The combination of 6 coupons with 5 as the greatest no. will be

$$\frac{\{1, 2, 3, 4, 5\} \times \{1, 2, 3, 4, 5\} \times \{1, 2, 3, 4, 5\} \times \text{---} \times \{1, 2, 3, 4, 5\} - \{1, 2, 3, 4\} \times \{1, 2, 3, 4\} \times \text{---} \times \{1, 2, 3, 4\}}{6 \text{ times} \quad \quad \quad 6 \text{ times}}$$

$$= 5^6 - 4^6$$

$$= 11,529 \quad \therefore \left[\frac{P}{1500} \right] = 7$$

15.(8) The available digits are 0, 1, 2, ..., 9

The first digit can be chosen in 9 ways

(0 not acceptable), the 2nd digit can be accepted in 9 ways (digits repetition not allowed)

Thus, the code can be made in $9 \times 9 = 81$ ways

Now, there are only 4 digits 1, 6, 8, 9 which can create confusion

Hence, the total number of codes which create confusion are $4 \times 3 = 12$

Out these 12 codes 69 & 96 will not create confusion. Hence, in total $12 - 2 = 10$ codes will create confusion

Hence, total codes without confusion are $81 - 10 = 71$

16.(0.20) The expression, $\frac{1}{11 \sin^2 \theta + 24 \sin \theta \cos \theta + 29 \cos^2 \theta}$ is essentially positive, therefore for the

expression to be maximum the denominator must be minimum.

$$\text{Let } a = 11 \sin^2 \theta + 24 \sin \theta \cos \theta + 29 \cos^2 \theta$$

$$= 20 + 9 \cos 2\theta + 12 \sin 2\theta = 20 \pm \sqrt{9^2 + 12^2} = 20 \pm 15 \begin{cases} \nearrow 35 \rightarrow \text{max value} \\ \searrow 5 \rightarrow \text{min value} \end{cases}$$

$$\therefore \text{Maximum value of given expression} = \frac{1}{5} = a$$

$$\begin{aligned}
 17.(1.77) \quad a &= \sum_{r=1}^{11} \tan^2\left(\frac{r\pi}{24}\right) = \tan^2\left(\frac{\pi}{24}\right) + \tan^2\left(\frac{2\pi}{24}\right) + \tan^2\left(\frac{3\pi}{24}\right) + \dots + \tan^2\left(\frac{11\pi}{24}\right) \\
 &= \left(\tan^2\frac{\pi}{24} + \cot^2\frac{\pi}{24}\right) + \dots + \left(\tan^2\frac{5\pi}{24} + \cot^2\frac{5\pi}{24}\right) + 1 = \left(\tan\frac{\pi}{24} + \cot\frac{\pi}{24}\right)^2 + \dots + \left(\tan\frac{5\pi}{24} + \cot\frac{5\pi}{24}\right)^2 - 9 \\
 &= 4 \left[\frac{1}{\sin^2\frac{\pi}{12}} + \dots + \frac{1}{\sin^2\frac{5\pi}{12}} \right] - 9 = 8 \left[\frac{1}{1-\cos\frac{\pi}{6}} + \frac{1}{1-\cos\frac{2\pi}{6}} + \frac{1}{1-\cos\frac{3\pi}{6}} + \frac{1}{1-\cos\frac{4\pi}{6}} + \frac{1}{1-\cos\frac{5\pi}{6}} \right] - 9 \\
 &= 8 \left[\frac{2}{2-\sqrt{3}} + 2 + 1 + \frac{2}{3} + \frac{2}{2+\sqrt{3}} \right] - 9 = \frac{253}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{Similarly} \quad b &= \tan^2\left(\frac{\pi}{24}\right) - \tan^2\left(\frac{2\pi}{24}\right) + \tan^2\left(\frac{3\pi}{24}\right) + \dots + \tan^2\left(\frac{11\pi}{24}\right) \\
 &= \left(\tan^2\frac{\pi}{24} + \cot^2\frac{\pi}{24}\right) - \left(\tan^2\frac{2\pi}{24} + \cot^2\frac{2\pi}{24}\right) + \left(\tan^2\frac{3\pi}{24} + \cot^2\frac{3\pi}{24}\right) - \left(\tan^2\frac{4\pi}{24} + \cot^2\frac{2\pi}{24}\right) \\
 &\quad + \left(\tan^2\frac{5\pi}{24} + \cot^2\frac{5\pi}{24}\right) - 1 \\
 &= \left(\tan\frac{\pi}{24} + \cot\frac{\pi}{24}\right)^2 - \left(\tan\frac{2\pi}{24} + \cot\frac{2\pi}{24}\right)^2 + \dots + \left(\tan\frac{5\pi}{24} + \cot\frac{5\pi}{24}\right)^2 - 3 \\
 &= 4 \left[\frac{1}{\sin^2\frac{\pi}{12}} - \frac{1}{\sin^2\frac{\pi}{12}} + \frac{1}{\sin^2\frac{3\pi}{12}} - \frac{1}{\sin^2\frac{4\pi}{12}} + \frac{1}{\sin^2\frac{5\pi}{12}} \right] - 3 \\
 &= 4 \left[\frac{2}{1-\cos\frac{\pi}{6}} - 4 + 2 - \frac{4}{3} + \frac{2}{1-\cos\frac{5\pi}{6}} \right] - 3 = 4 \left[16 - \frac{10}{3} \right] - 3 = \frac{143}{3}
 \end{aligned}$$

18.(495) Total number of ways to select four seats from 15 such that no two of them are consecutive is the total number of integral solutions of $x_1 + x_2 + x_3 + x_4 + x_5 = 11$

Under the condition that $x_1 \geq 0, x_2, x_3, x_4 \geq 1, x_5 \geq 0 = {}^{12}C_4$